

Load-Aware A* Path Planning for Quadruped Robots

Day 25 of 30 · · Week 4: Results Review & Integration · Member 1 — Results Section Input

NAME: HARINI P (Department of Computer Science and Engineering)

1. Nature of Work Done

Day 25 transitions Member 1’s role from audit and review to original writing. The primary deliverable is the first complete draft of Section VI (Discussion) — the paper’s most analytically dense section. Section VI must synthesise the algorithm design, the theoretical framework, and the 20-scenario results into a coherent narrative that explains not just what Load-Aware A* achieves, but why it achieves it, grounded in physics, graph search theory, and stability mechanics.

Task	Description	Deliverable	Status
Task 1	Begin writing Discussion section	Section VI first draft — all 5 sub-sections	Complete ✓
Task 2	Explain why Load-Aware A* performs better	VI-A: Performance superiority analysis with causal chain	Complete ✓
Task 3	Theoretical justification	VI-B: Formal theoretical grounding (graph theory, stability mechanics, optimality)	Complete ✓
Task 4	Explain stability benefits	VI-C through VI-E: Stability benefit analysis, payload scalability, limitations	Complete ✓

PART A — SECTION VI: DISCUSSION (First Draft)

The following is the first complete draft of Section VI as it will appear in the paper. Member 4 will review and merge with Section V results before final submission.

VI. DISCUSSION

This section analyses the Load-Aware A* results in the context of the design objectives stated in Section III-A: to produce a path that is simultaneously geometrically efficient and mechanically safe for a payload-carrying quadruped robot. The discussion proceeds from performance analysis, through theoretical justification, to stability benefit quantification, scalability assessment, and an honest examination of limitations. Throughout, we distinguish between results that are guaranteed by the algorithm’s mathematical structure and those that are empirically observed across the 20 test scenarios.

VI-A. Why Load-Aware A* Performs Better Than Standard A*

The performance advantage of Load-Aware A* over Standard A* is not coincidental — it is the direct consequence of an information asymmetry between the two algorithms. Standard A* is blind to the robot’s physical state: it treats every free cell as an identical unit of traversal cost, with no knowledge of payload mass, foot geometry, or CoM displacement. Load-Aware A*, by contrast, encodes the robot’s

physical configuration directly into the cost function via the stability penalty $S(n)$. This structural difference has four specific performance consequences.

VI-A.1 Consequence 1: Informed Routing Through the Stability Landscape

Standard A* computes the globally shortest path in terms of step count. When this shortest path happens to pass through cells where the payload's CoM displacement is large — that is, cells in the interior of the grid near the payload position — Standard A* has no mechanism to detect or avoid them. It will traverse these cells without penalty, producing a geometrically optimal but mechanically suboptimal trajectory.

Load-Aware A*, by contrast, has access to the full stability landscape via S_{map} (Equation 12). Before any node is expanded, every free cell already has a pre-computed stability cost. The modified evaluation function $f(n) = g(n) + \alpha \cdot S(n) + h(n)$ (Equation 10) therefore propagates stability awareness through the entire search tree: nodes with high S values accumulate larger g -costs and become less preferred, naturally routing the algorithm toward cells with lower CoM displacement even if those cells require a slightly longer path.

The key insight is that this routing is global — it reflects the stability cost of the entire path from start to goal, not just the next cell. This is the defining advantage of integrating stability into the search objective rather than applying it as a post-processing filter.

VI-A.2 Consequence 2: The S_{avg} Reduction is Structurally Guaranteed

For any scenario where the two algorithms produce different paths ($D = \text{True}$), it can be proven that Load-Aware A* returns a path with lower or equal total stability cost. This is not an empirical observation but a direct consequence of optimality: Load-Aware A* returns the path with the minimum value of $C = \sum(1 + \alpha \cdot S(n))$ over all nodes in the path. If a lower- S path exists, the algorithm will find it. If it does not (because all routes pass through equally unstable terrain), the two algorithms may find the same path.

This structural guarantee means that every scenario with $D = \text{True}$ in the results table is one where a lower-stability-cost path existed and was found. The S_{avg} reduction observed in the results is therefore not a statistical artefact of the test scenarios — it is the algorithm operating exactly as designed.

VI-A.3 Consequence 3: The dL Cost is the Price of Stability Information

When Load-Aware A* produces a path with $dL > 0$ (extra steps compared to Standard A*), this represents the algorithm trading geometric efficiency for stability. This trade-off is rational under the following condition: the gain in stability (ΔS_{avg} or ΔM_{min}) must justify the additional step cost.

In the 20-scenario results, dL is constrained to 1–3 steps on a 10×10 grid where the minimum path is 18 steps. An extra step represents at most a 5.6% increase in geometric path length. When this cost is accompanied by a substantial reduction in S_{avg} or an M_{min} improvement that crosses the 0.05 m safety threshold, the trade-off is strongly justified: the robot travels a marginally longer path but does so without approaching the tip-over boundary.

It is important to note that $dL > 0$ does not indicate suboptimality. Load-Aware A* is optimal with respect to its composite objective $f(n) = g(n) + \alpha \cdot S(n) + h(n)$. The apparent 'extra steps' are extra steps only when measured against the purely geometric objective of Standard A*, which is a different — and, for payload-carrying robots, physically inadequate — cost criterion.

VI-A.4 Consequence 4: Negligible Computational Overhead

A potential objection to stability-aware planning is computational cost: does embedding stability into the search make the algorithm too slow for practical use? The results reject this objection. The $dT\%$ values across all 20 scenarios remain below 5%, confirming that the per-node overhead introduced by Load-Aware A* is negligible.

This result is architecturally explained by Equation 12: S_{map} is computed once in $O(R \times C) = O(100)$ before the search begins. During the search, each $S(n)$ lookup is $O(1)$ — a single Python dictionary access. The stability penalty therefore adds no asymptotic complexity to the search and only a trivial constant factor per node expansion. The algorithm's time complexity remains identical to Standard A* in practice.

VI-B. Theoretical Justification

VI-B.1 Graph-Theoretic Foundation

Load-Aware A* is a weighted shortest-path algorithm operating on a directed graph $G = (V, E)$ where V is the set of free cells and E is the set of 4-connected edges between adjacent free cells. The edge weight $w(n, n') = 1 + \alpha \cdot S(n')$ assigns a combined geometric-stability cost to each traversal. Under this formulation, the algorithm's optimality follows directly from the classical A* optimality theorem (Hart, Nilsson, and Raphael, 1968):

A* returns an optimal path if and only if the heuristic $h(n)$ is admissible: $h(n) \leq h^*(n)$ for all nodes n , where $h^*(n)$ is the true minimum cost from n to the goal.

In Load-Aware A*, $h(n) = |r - r_g| + |c - c_g|$ (Manhattan distance, Equation 8). The true minimum cost from n to the goal under the combined objective is at least the number of remaining steps, since each step costs at least 1 (the base step cost). The stability penalty $\alpha \cdot S(n)$ is always non-negative and is not included in $h(n)$. Therefore $h(n) \leq h^*(n)$ holds for all n under the composite cost function, and Load-Aware A* is admissible.

Consistency (monotonicity) is also preserved: for any two adjacent nodes n and n' , $h(n) \leq w(n, n') + h(n')$ reduces to the triangle inequality for Manhattan distance, which holds by construction for 4-connected grids. Consistency implies that each node is expanded at most once, giving the algorithm the same $O((V + E) \log V)$ complexity bound as Standard A*.

VI-B.2 Physical Mechanics Justification

The stability penalty $S(n) = \|x_{\text{CoM}}(n) - x_c(n)\|_2$ (Equation 5) has a direct physical interpretation rooted in quasi-static stability theory for legged robots. For a robot in quasi-static motion (assumption A4 in Methodology v2.0), the risk of tip-over at a given cell is determined by how far the composite Centre of Mass lies from the centroid of the support polygon. A CoM displacement of zero ($S(n) = 0$) means the robot is perfectly balanced; increasing $S(n)$ means the CoM is moving toward a polygon edge, reducing the stability margin $M(n)$.

The relationship between $S(n)$ and $M(n)$ is monotonic for a given foot configuration: as $S(n)$ increases (CoM moves further from centroid), $M(n)$ decreases (CoM approaches an edge). This monotonicity justifies using $S(n)$ as a proxy for $M(n)$ in the cost function: minimising accumulated $S(n)$ along a path is mathematically equivalent to maximising the cumulative distance of the CoM from the polygon boundary, which is the operational definition of mechanical stability in quasi-static locomotion.

The safety threshold $M(n) > 0.05 \text{ m}$ (5 cm) is drawn from the quadruped locomotion literature as the minimum margin required to absorb small perturbations (uneven terrain, sensor latency, actuator delay) without triggering a dynamic instability. By routing through cells where $S(n)$ is low, Load-Aware A* preferentially maintains $M(n)$ above this threshold throughout the path — not just at the start and goal.

VI-B.3 Justification for Continuous Penalty over Binary Constraint

A theoretically simpler approach to stability-aware planning would be to treat cells where $M(n) \leq 0.05 \text{ m}$ as obstacles — a binary constraint. This approach fails on three theoretical grounds:

- **Completeness loss:** In high-density obstacle scenarios, all paths from start to goal may pass through at least one cell with $M(n) \leq 0.05 \text{ m}$. Binary exclusion would cause planning failure even

when a physically traversable path (albeit marginally unstable) exists. Load-Aware A*'s continuous penalty preserves completeness: the algorithm always returns the best available path.

- Non-differentiability of the stability objective: A binary constraint introduces a discontinuous cost landscape, where a cell is either 'free' or 'blocked' with no gradation. The continuous $S(n)$ penalty preserves a smooth cost gradient that guides the search progressively toward stable regions, allowing the algorithm to find near-stable paths even when the ideal path is unavailable.
- Loss of worst-case guarantees: Binary exclusion eliminates cells globally, regardless of whether a small or large margin violation is occurring. A cell with $M(n) = 0.049$ m is treated identically to a cell with $M(n) = -0.3$ m (severe tip-over risk), even though the former may be safely traversable with acceptable perturbation tolerance. The continuous $S(n)$ distinguishes these cases and assigns proportionally higher cost to more dangerous cells.

VI-B.4 Admissibility Proof (Formal Statement)

Theorem: The Manhattan heuristic $h(n)$ is admissible for Load-Aware A* under the combined cost function $f(n) = g(n) + \alpha \cdot S(n) + h(n)$.

Proof: Let $h^*(n)$ be the true minimum cost from n to the goal under the combined objective. The minimum cost of any remaining path from n to goal is bounded below by the minimum number of steps required, since: (a) each step contributes at least 1 to g , and (b) $\alpha \cdot S(n) \geq 0$ for all n ($S(n)$ is a Euclidean distance and is non-negative). The minimum number of steps from n to the goal on a 4-connected grid is the Manhattan distance $h(n) = |r - r_g| + |c - c_g|$. Therefore $h(n) \leq h^*(n)$ for all n , and the heuristic is admissible. ■

VI-C. Stability Benefits — Quantitative Analysis

VI-C.1 The Two Stability Metrics and Their Roles

The results table records two complementary stability metrics: S_{avg} (mean stability penalty over the path) and M_{min} (minimum stability margin over the path). These metrics measure different aspects of the stability benefit and must both be reported to characterise Load-Aware A*'s performance fully.

Metric	What It Measures	Why It Matters	Expected Load-Aware Improvement
S_{avg}	Average CoM displacement from polygon centroid across all path cells	Path-level stability aggregate; lower = more stable journey overall	STRONGER than Standard A* across all D=True scenarios (structural guarantee)
M_{min}	Worst-case safety margin: minimum perpendicular CoM-to-edge distance over path	Single-cell safety floor; one $M_{min} \leq 0.05$ m invalidates the path for safe traversal	STRONGER than Standard A*: $M_{min}(LA) > 0.05$ m where $M_{min}(Std) \leq 0.05$ m

The relationship between the two metrics defines four possible scenario outcomes. Understanding which quadrant each scenario falls into is critical for interpreting the results table:

Quadrant	$S_{avg}(LA)$ vs Std	$M_{min}(LA)$ vs Std	Interpretation
Q1 — Full Improvement	Lower	Higher (above 0.05 m)	Load-Aware A* improves both average stability and worst-case safety — strongest result

Quadrant	S_avg(LA) vs Std	M_min(LA) vs Std	Interpretation
Q2 — Safety Improvement Only	Marginal difference	Higher (crosses 0.05 m threshold)	Same average trajectory quality; single dangerous cell eliminated — critical safety gain
Q3 — Average Improvement Only	Lower	Both above threshold	Both algorithms safe; Load-Aware smoother cumulative stability profile
Q4 — Baseline Equal (D = False)	Equal	Equal	Same path found; mp = 3 kg or payload outside routing conflict zone

VI-C.2 Stability Benefit Across Payload Bands

The magnitude of the stability benefit scales with payload mass because heavier payloads produce larger CoM displacement (Equation 4). As $m_{\square}$ increases from 3 kg to 9 kg, the x_{CoM} shifts further from the foot centroid x_b , producing larger $S(n)$ values throughout the grid interior. This creates a stronger incentive for Load-Aware A* to deviate from the geometric shortest path, resulting in larger ΔS_{avg} and ΔM_{min} at higher payload masses.

Payload Band	CoM Shift Magnitude	Expected Stability Benefit	D Rate	Dominant Quadrant
$m_{\square} = 3$ kg (30% body mass)	Small: x_{CoM} displaced ≈ 0.07 m from centroid	MODERATE: marginal improvement; $S(n)$ gradient weak	Low ($\approx 20\%$)	Q4 (baseline equal) or Q3
$m_{\square} = 5$ kg (50% body mass)	Moderate: x_{CoM} displaced ≈ 0.12 m from centroid	STRONGER: clear S_{avg} reduction; M_{min} may cross threshold	Moderate ($\approx 60\%$)	Q1 or Q2
$m_{\square} = 7$ kg (70% body mass)	Large: x_{CoM} displaced ≈ 0.17 m from centroid	STRONGER: consistent improvement; multiple unsafe cells avoided	High ($\approx 90\%$)	Q1 (dominant)
$m_{\square} = 9$ kg (90% body mass)	Very large: x_{CoM} displaced ≈ 0.21 m from centroid	STRONGER: maximum benefit; Standard A* path frequently violates M threshold	Very high ($\approx 100\%$)	Q1 (dominant)

VI-C.3 M_{min} as the Critical Safety Metric

While S_{avg} is the primary effectiveness metric (it measures the algorithm's ability to reduce cumulative instability), M_{min} is the critical safety metric for the following reason: a path in which even a single cell has $M(n) \leq 0.05$ m exposes the robot to a tip-over event regardless of how stable the rest of the path is. M_{min} is therefore a go/no-go criterion, not a continuous quality score.

The most important result in the entire paper is the comparison of M_{min} values between Standard A* and Load-Aware A* at $m_{\square} \geq 5$ kg. When $M_{\text{min}}(\text{Std}) \leq 0.05$ m and $M_{\text{min}}(\text{LA}) > 0.05$ m in the same scenario, Load-Aware A* has not merely improved a metric — it has produced a path that is safe where Standard A* produced a path that was mechanically dangerous. This binary safe/unsafe distinction is

what makes Load-Aware A* a practically significant contribution, not merely an incremental improvement in an existing algorithm.

VI-D. Payload Scalability and Parameter Robustness

A practical limitation of many stability-aware planning methods is parameter sensitivity: algorithms require re-tuning as the robot's physical configuration changes. Load-Aware A* is designed to eliminate this limitation through two architectural properties.

VI-D.1 Automatic Adaptation via the Physical Model

When the payload mass $m_{\square}$ changes, the stability map S_{map} is automatically recomputed from Equation 4 before the next search. The algorithm does not require any manual adjustment: the same weight $\alpha = 1.0$ produces appropriately scaled stability penalties across all payload levels because the penalty $S(n)$ itself encodes the physical magnitude of the CoM displacement. A heavier payload at the same position produces a larger $S(n)$ at surrounding cells, which is automatically reflected in the cost function and the resulting path.

This is fundamentally different from methods that use fixed penalty zones (e.g. 'avoid cells within 2 metres of the payload') which require manual re-calibration when payload position or mass changes. Load-Aware A* re-calibrates automatically via the physics of Equation 4.

VI-D.2 Alpha Robustness

The stability weight $\alpha = 1.0$ was selected as the minimum value producing consistent path divergence at $m_{\square} \geq 5$ kg. This selection criterion has an important implication: the algorithm is not over-tuned. A smaller α would under-penalise instability and fail to route around dangerous cells; a larger α would over-penalise instability and produce unnecessarily long detours. At $\alpha = 1.0$, the algorithm is calibrated at the minimum effective sensitivity — making it robust to small perturbations in the environment without inducing unnecessary path elongation.

Property	Standard A*	Load-Aware A* ($\alpha=1.0$)	Advantage
$m_{\square}$ change (e.g. 5 kg $\rightarrow$ 7 kg)	No re-computation needed (stability ignored)	S_{map} recomputed automatically via Eq. 4	ADVANTAGE: physical adaptation without parameter change
Payload position change	No re-computation needed (stability ignored)	S_{map} recomputed; new spatial gradient	ADVANTAGE: spatial routing updates automatically
Grid size change	Scales in $O(V \log V)$	Scales identically; S_{map} $O(R \times C)$ pre-compute	ADVANTAGE: no additional scaling cost
New obstacle placement	New path search required	New path search + S_{map} update	MODERATE: S_{map} adds $O(R \times C)$ to re-planning cost

VI-E. Limitations and Scope Boundaries

A credible Discussion section must acknowledge the conditions under which the algorithm's advantages do not hold or may be reduced. The following limitations are inherent to the current formulation and should be stated explicitly in the paper.

- Quasi-static assumption: The stability model (Equations 5–6) assumes the robot is in quasi-static equilibrium at each cell — it does not account for inertial forces during locomotion.

At high speeds, dynamic effects (centrifugal force, inertial CoM shift) may invalidate the static $S(n)$ values. The algorithm is therefore most applicable to slow-to-moderate speed locomotion.

- 2D grid abstraction: The environment is modelled as a discrete 2D grid with binary occupancy. Real terrain has continuous geometry, varying friction, and 3D slope — none of which are captured in the current $S(n)$ model. Extension to 3D height maps or continuous-space planners is future work.
- Single payload assumption: The model assumes one rigid payload at a fixed position x_{payload} . Multiple payloads, shifting payloads (e.g. liquid containers), or dynamically moving loads would require an extended CoM model.
- Foot contact assumption: The support polygon is computed from four fixed foot positions (Equation 3) assuming all four feet are always in contact. During gait transitions (e.g. trot gait with alternating foot lifts), the support polygon changes dynamically. The current model is a static approximation valid for standing or slow-walking gaits only.
- Alpha generalisability: The weight $\alpha = 1.0$ was validated on a 10×10 grid with specific obstacle densities and payload ranges. For substantially different grid scales, obstacle distributions, or payload ranges, re-tuning of α may be required.

Limitation framing guidance: These limitations should be presented in the paper as scope boundaries, not algorithmic failures. Each limitation identifies the conditions under which Load-Aware A^* as presented here applies, and opens a clear direction for future work. The quasi-static limitation in particular is standard in the quadruped locomotion literature and does not diminish the contribution within its stated scope.

PART B — MEMBER NOTES & INTEGRATION INSTRUCTIONS

2. Discussion Section Structure — Integration Map

The five Discussion sub-sections drafted above map to the following paper positions and connect to the adjacent sections as indicated. Member 4 should use this map when merging the Discussion draft with the completed results.

Sub-section	Core Argument	Connects To	Key Evidence Needed From Section V
VI-A: Why better	Information asymmetry: LA A^* has stability landscape; Std A^* does not	Sections IV (algorithm design) $\rightarrow$ VI-A $\rightarrow$ Conclusion	$S_{\text{avg}}(\text{LA}) < S_{\text{avg}}(\text{Std})$ for all $D=\text{True}$ scenarios
VI-B: Theory	Admissibility proof + physical mechanics justification	Section III (methodology) $\rightarrow$ VI-B	No numerical results needed; purely theoretical
VI-C: Stability benefits	S_{avg} and M_{min} improvement quantified; M_{min} as go/no-go criterion	Section V results $\rightarrow$ VI-C	$M_{\text{min}}(\text{LA}) > 0.05 \text{ m}$ vs $M_{\text{min}}(\text{Std}) \leq 0.05 \text{ m}$ count
VI-D: Scalability	Automatic adaptation via physics; α robustness	Section III-F (parameters) $\rightarrow$ VI-D	D rate by payload band; dL by payload band
VI-E: Limitations	Quasi-static, 2D grid, single payload, foot contact	VI-E $\rightarrow$ Future Work / Conclusion	No numerical results; scope boundary statement

3. Key Sentences for Abstract and Conclusion

The following sentences, drawn directly from the Discussion draft, are recommended for use in the Abstract and Conclusion sections:

3.1 For the Abstract

- 'Load-Aware A* embeds a pre-computed stability penalty map into the A* cost function, routing the robot through cells with lower Centre-of-Mass displacement without sacrificing heuristic admissibility or introducing measurable computational overhead.'
- 'Across 20 test scenarios, Load-Aware A* consistently achieves lower mean stability penalty (S_{avg}) and higher minimum stability margin (M_{min}) than Standard A*, with path length increases of 1–3 steps and compute overhead below 5%.'
- 'The algorithm scales automatically across payload masses of 3–9 kg without parameter re-tuning, as the physical stability model adapts to the payload configuration at each planning call.'

3.2 For the Conclusion

- 'The principal contribution of this work is the demonstration that stability-aware global path planning can be achieved within the A* framework at negligible computational cost, by pre-computing a stability map and integrating it as a continuous additive penalty in the search objective.'
- 'The most significant safety result is the consistent achievement of $M_{min} > 0.05$ m on Load-Aware paths in scenarios where Standard A* produces paths with $M_{min} \leq 0.05$ m, demonstrating a transition from mechanically unsafe to safe trajectory planning under payload conditions.'
- 'Future work should address the quasi-static limitation by incorporating dynamic CoM models, and extend the framework to 3D terrain with variable foot contact polygons.'